

5(a)(i)	thylakoid (membranes) / grana / lamellae ;	1
5(a)(ii)	reduced NADP / ATP / oxygen ; R reduced NAD	1
5(a)(iii)	<i>any four from:</i> 1 <i>ref. to</i> rubisco ; 2 forms unstable 6C compound ; 3 (splits into) two molecules of, GP / glycerate-(3)-phosphate ; 4 GP reduced to, TP / triose phosphate ; 5 using reduced NADP and ATP ; 6 TP(s) used to form glucose ; Ignore hexose 7 <i>ref. to</i> polymerisation / condensation / formation of glycosidic bonds ;	4
5(b)(i)	more carbon dioxide react with RuBP / more CO₂ fixation / more carbon dioxide available to bind to rubisco ; ora more, Calvin cycle / light independent reaction / detail ; ora	2
5(b)(ii)	<i>any three from:</i> 1 as day length increases rate (of photosynthesis) increases / positive correlation ; 2 data quote: two day lengths in hours and, two carbon dioxide concentrations / difference between two carbon dioxide concentrations ; 3 more light is absorbed by, photosystems / pigments / named pigment ; 4 <i>ref. to</i> more, photophosphorylation / light dependent reaction / photolysis ; 5 more, oxygen / reduced NADP / ATP / glucose / starch, produced ;	3

Question	Answer	Marks
5(c)(i)	(pH 8.4) optimum pH of, rubisco / enzymes ; fewer, H ⁺ / protons, result in higher rate of, photosynthesis / activity of enzymes ;	2
5(c)(ii)	<i>idea of</i> less carbon dioxide, absorbed / fixed (so ocean acidification not reduced) ;	1